25900688

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

THIRD SEMESTER

Discipline Specific Elective (DSE) - MG3DSEZGYA00 - BIOLOGY OF GENOMES

(2024 ADMISSION ONWARDS)

Scheme for Valuation

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

(Multiple Choice Questions; Answer all questions; Each question carries 1 mark)

1. Compared with animals and plants, many fungal nuclear genomes are: [U] / [CO1]
Ans: b) Typically compact and gene-dense, with fewer introns on average

2. The Arabidopsis thaliana genome is best described as: [K] / [CO1]
Ans: b) Diploid with 5 chromosomes

3. Which organelle is thought to have originated from an endosymbiotic event with a photosynthetic cyanobacterium? [K] / [CO1]
Ans: c) Chloroplast

4. Which process allows plasmids to move directly from one bacterial cell to another? [U] / [CO2]
Ans: b) Conjugation

5. Which of the following statements about bacterial chromosome organization is correct? [A] / [CO2]
Ans: b) Supercoiling and NAP interactions condense the chromosome

6. Why was sequencing the SARS-CoV-2 genome important for vaccine development? [E] / [CO3]
Ans: a) It helped identify the spike protein as a target

7. What does the "+ssRNA" genome of Tobacco Mosaic Virus mean? [E] / [CO3]
Ans: b) It can directly function as mRNA for protein synthesis

8. Why is Bayesian inference increasingly used in phylogenetics? [An] /[CO4]
Ans: b) It uses statistical probability to estimate tree confidence

9. How is the Geological Time Scale constructed? [An]
Ans: b) By radiometric dating and fossil records **/[CO4]**

10. The total chromosome number in chimpanzees is: [K] /
Ans: d) 48 **[CO5]**

[1x10 = 10]

Part B

(Short Answer Questions; Answer any 14 questions; Each question carries 2 marks)

11. What percentage of the human genome is estimated to be protein-coding? [K] /
Ans: Approximately **1.5% to 2%**. **[CO1]**
[**Valuation:** (2 marks) Award full marks for any value within the 1% to 3% range.]

12. How do mitochondrial ribosomes link mitochondria to bacterial ancestors? [U] /
Ans: Mitochondrial ribosomes are of the **70S** type, similar in size and structure to **prokaryotic (bacterial) ribosomes**, rather than the 80S ribosomes found in the eukaryotic cytoplasm. **[CO1]**

[**Valuation:** (1 mark) for mentioning 70S type. (1 mark) for linking this to prokaryotic/bacterial ribosomes.]

13. Like mitochondria, chloroplasts are also believed to have an endosymbiotic origin. Mention its probable ancestor? [K] / [CO1]
Ans: A **photosynthetic cyanobacterium** (or simply **cyanobacterium**).
[**Valuation:** (2 marks) for the correct term.]

14. Explain the difference between tandem repeats and interspersed repeats. [U] /
Ans: **Tandem repeats:** DNA sequences repeated one after another in a **head-to-tail** fashion (e.g., AAT-AAT-AAT). **[CO1]**

Interspersed repeats: DNA sequences (like transposons/retrotransposons) that are **scattered** at different locations throughout the genome.

[**Valuation:** (1 mark) for the correct definition of tandem repeats. (1 mark) for the correct definition of interspersed repeats.]

15. Explain the concept of integrons and analyze their role in the development and spread of antibiotic resistance in bacteria. [U] / [CO2]
Ans: **Concept:** Integrons are genetic elements that can **capture and express gene cassettes**.
Role: They act as platforms to **acquire and assemble multiple antibiotic resistance genes** (as cassettes), allowing bacteria to quickly become multi-drug resistant.
[Valuation: (1 mark) for the concept (capturing/expressing gene cassettes). (1 mark) for the role (acquiring/stacking resistance genes).]

16. How does having minimal non-coding DNA help prokaryotes adapt to their environment? [A] / [CO2]

Ans: It allows for a **compact genome**, which enables **rapid replication** and cell division. This metabolic efficiency (less energy spent on non-essentials) allows them to multiply quickly and exploit favourable conditions. This energy conservation is a major competitive advantage, especially in resource-poor environments where they must compete with other microbes for a limited supply of nutrients.

[**Valuation:** (2 marks) for linking minimal non-coding DNA to rapid replication/metabolic efficiency.]

17. Describe the structure and composition of the bacterial chromosome found in the nucleoid. [U] / [CO2]

Ans: It is typically a **single, circular, double-stranded DNA** molecule. It is located in the **nucleoid** (a non-membrane-bound region) and is highly compacted by **supercoiling** and **Nucleoid-Associated Proteins (NAPs)**.

[**Valuation:** (1 mark) for "single, circular, dsDNA." (1 mark) for "nucleoid/supercoiling/NAPs.]

18. Mention two general features of prokaryotic genomes. [K] / [CO2]

Ans: Any two of the following:

- Mostly **circular**
- **Haploid**
- **High gene density** (little non-coding DNA)
- Often contain **plasmids**
- Genes often organized into **operons**

[**Valuation:** (1 mark) for each correct feature (max 2 marks).]

19. Compare single-stranded and double-stranded viral genomes and evaluate their stability inside the host. [E] / [CO3]

Ans: **Double-stranded (ds)** genomes (especially dsDNA) are generally **more stable** and less prone to degradation by exonucleases. **Single-stranded (ss)** genomes (especially ssRNA) are **less stable** and more susceptible to host enzymes and replication errors.

[**Valuation:** (1 mark) for identifying ds genomes as more stable. (1 mark) for identifying ss genomes as less stable.]

20. How does evaluating variants of SARS-CoV-2 help in deciding monoclonal antibody use? [E] / [CO3]

Ans: Monoclonal antibodies target specific sites (epitopes) on the **spike protein**. Evaluating variants shows if **mutations** have occurred in these target sites, which could make the antibody **ineffective** (unable to bind).

[**Valuation:** (1 mark) for mentioning mutations in the target/spike protein. (1 mark) for linking this to potential antibody ineffectiveness.]

21. What makes the SARS-CoV-2 genome suitable for rapid mutation and evolution? [E] / [CO3]

Ans: It is an **RNA virus**. Its replication enzyme, **RNA-dependent RNA polymerase (RdRp)**, **lacks a strong proofreading mechanism**, leading to a high error/mutation rate.

[**Valuation:** (1 mark) for "RNA virus." (1 mark) for "lack of proofreading/high error rate of RdRp."]

22. **Which enzyme is used by SARS-CoV-2 to copy its RNA genome?** **[K]** /
Ans: **RNA-dependent RNA polymerase (RdRp)**. **[CO3]**
[Valuation: (2 marks) for the correct name or acronym.]

23. **What is a haplotype block?** **[U]** /
Ans: A region of a chromosome containing a set of **linked alleles (a haplotype)** that are **inherited together** as a unit. This is because there is **low recombination** within the block. **[CO4]**
[**Valuation:** (1 mark) for "linked alleles/haplotype inherited together." (1 mark) for "low recombination."]

24. **State an application of molecular clocks.** **[A]** /
Ans: To **estimate the time of divergence** between two species or lineages. (Also accept: "dating evolutionary events"). **[CO4]**
[Valuation: (2 marks) for the correct application.]

25. **Define a phylogram.** **[K]** /
Ans: A type of **phylogenetic tree** where the **branch lengths are proportional** to the amount of **evolutionary change** (or genetic distance) that has occurred. **[CO4]**
[**Valuation:** (1 mark) for "phylogenetic tree." (1 mark) for "branch lengths proportional to evolutionary change."]

26. **Explain the difference between rooted and unrooted phylogenetic trees.** **[U]** /
Ans: **Rooted tree:** Has a **common ancestor (the root)** for all entities in the tree, showing the **direction of evolution**. **[CO4]**
Unrooted tree: Shows the **relationships** (relatedness) between entities but does **not** specify the common ancestor or the evolutionary path.

[**Valuation:** (1 mark) for the correct definition of rooted. (1 mark) for the correct definition of unrooted.]

[2x14 = 28]

Part C

(Short Essay Questions; Answer any 8 questions; Each question carries 4 marks)

27. **Discuss how histone proteins contribute to the packaging of DNA in the human genome** **[U]** **[CO1]** /

Ans:

- DNA is **negatively charged** (due to phosphate groups). **(1 mark)**
- Histones are **positively charged** proteins (rich in lysine and arginine). **(1 mark)**
- Eight histones (two each of H2A, H2B, H3, H4) form a protein core called an **octamer**. **(1 mark)**
- The negatively charged DNA wraps around this positive octamer core to form a **nucleosome**, which is the basic unit of chromatin, leading to significant compaction. **(1 mark)**

[**Valuation:** Award 1 mark for each key point.]

28. **Briefly list at least two significant differences between the structure or organization of the mitochondrial genome and the nuclear genome of eukaryotic cells** **[U] [CO1]** /

Ans: Any two of the following, explained clearly:

- **Structure:** Mitochondrial (mt) genome is **circular**, while the nuclear genome consists of **linear** chromosomes.
- **Size:** mt genome is very **small** (~16.5 kb in humans), while the nuclear genome is very **large** (~3.2 Gb).
- **Gene Density:** mt genome is **gene-dense** with very few introns, while the nuclear genome is mostly **non-coding** with many introns.
- **Packaging:** mt genome **lacks histones**, while the nuclear genome is packaged by histones into chromatin.
- **Inheritance:** mt genome is typically **maternally inherited**, while the nuclear genome is inherited from **both parents**.

[Valuation: (2 marks) for the first clear comparative point. (2 marks) for the second clear comparative point. (1 mark per point if only listed without comparison).]

29. **Briefly explain the concept of "jumping genes" in maize and name two distinct types of transposable elements found in this plant.** **[U] [CO1]** /

Ans: **Concept:** "Jumping genes" (or transposable elements, TEs) are DNA sequences that can **move (transpose) from one location in the genome to another**. (1 mark). Their movement can cause mutations, alter gene expression, and was first discovered by Barbara McClintock in maize. (1 mark).

Types: The **Ac (Activator)** element (1 mark) and the **Ds (Dissociation)** element (1 mark). (Or accept Class I/Retrotransposons and Class II/DNA transposons).

[**Valuation:** (2 marks) for a clear explanation of the concept. (2 marks) for naming the two types (1 mark each)]

30. **Using the lac operon as an example, outline its structure and explain how it regulates gene expression in response to the presence or absence of lactose.** **[A] [CO2]** /

Ans: **Structure:** (1 mark) Composed of a **Promoter (P)**, **Operator (O)**, and three structural genes (**lacZ, lacY, lacA**). A separate regulator gene (**lacI**) codes for a repressor protein.

Lactose: (1 mark) The **repressor protein binds to the operator**, physically blocking RNA polymerase from transcribing the structural genes (genes are **OFF**).

Lactose Present: (2 marks) **Allolactose** (an inducer derived from lactose) **binds to the repressor protein** (1 mark). This changes the repressor's shape, causing it to **detach from the operator**. RNA polymerase can then transcribe the genes (genes are **ON**) (1 mark).

[**Valuation:** As per the mark breakdown.]

31. **Explain the structure of *E. coli* genome with a simple diagram.** **[U] [CO2]** /

Ans:

Diagram: (2 marks) A **simple, clear circle** (1 mark). Must be labelled with at least **'circular DNA'**, **'oriC'** (origin of replication), and **'ter'** (terminus) (1 mark for labels).

Explanation: (2 marks) The *E. coli* genome is a **single, circular, double-stranded DNA** molecule (1 mark). It is located in the **nucleoid** and is compacted by **supercoiling** and NAPs. It is highly **gene-dense** with genes often grouped into operons (1 mark).

[**Valuation:** Award marks for a correct diagram and a conceptually sound explanation.]

32. Analyze why SV40 is widely used as a model organism in molecular biology and cancer research. [An] / [CO3]

Ans: It has a **very small, simple genome** (circular dsDNA, ~5.2 kb) that is easy to manipulate. **(1 mark)**

It **replicates in eukaryotic cells** (monkey and human) and **uses the host cell's machinery** for DNA replication, transcription, and translation, making it ideal for studying these eukaryotic processes. (1 mark)

It contains powerful **promoters** used in expression vectors. **(1 mark)**

It is a **tumor virus**; its **Large T-antigen** is an **oncogene** that can **transform** cells, providing a simple model to study cell cycle control and cancer. **(1 mark)**

[**Valuation:** (1 mark) for each distinct, relevant point.]

33. What is the role of the membrane (M) protein in the virus? [U] / [CO3]

Ans: (e.g., in Coronaviruses)

- It is the **most abundant** structural protein. **(1 mark)**
- It is a transmembrane glycoprotein that **defines the shape and curvature** of the viral **envelope**. **(1 mark)**
- It plays a crucial role in **viral assembly** and **budding** from the host cell. **(1 mark)**
- It interacts with other structural proteins (like E, S, and N) to **stabilize the nucleocapsid** and organize virion formation. **(1 mark)**

[**Valuation:** (1 mark) for each correct role/feature.]

34. Discuss the role of cladistics in constructing cladograms. [U] / [CO4]

Ans: **Cladistics:** (2 marks) A method of phylogenetic classification that groups organisms based on **shared derived characteristics (synapomorphies)** (1 mark). It assumes that when a new trait evolves, it is passed on to all descendants (1 mark).

Cladogram: (2 marks) A cladogram is the **diagram** resulting from a cladistic analysis (1 mark). It **illustrates the branching patterns** (topology) of evolutionary relationships, showing which groups (clades) share a more recent common ancestor based on those synapomorphies (1 mark).

[**Valuation: Liberal valuation]**

35. Explain how comparative genomics can reveal evolutionary divergence among species. [An] /[CO4]

Ans: Core Concept of Comparison:

Comparative genomics involves comparing the complete genomes ("instruction books") of different species. Evolutionary divergence is revealed by analyzing the **similarities and differences** in the DNA code. **(1 Mark)**

Role of Homology (Orthologs):

By comparing **homologous sequences** or **orthologs** (genes in different species inherited from a single ancestral gene), scientists can track divergence. The *number of differences* (mutations) that have accumulated between the orthologs (like human and chimp myoglobin) indicates how much the species have diverged since they split from that common ancestor. **(1 Mark)**

Role of Gene Duplication (Paralogs):

For explaining *paralogs* as a *mechanism* for divergence. Gene duplication creates **paralogs** (related genes within one species). This allows one copy to keep the original job, while the second copy is "free" to mutate and evolve a **new function**. This creation of new functions (e.g., alpha- and beta-globin) is a major way species diverge and develop new traits. **(1 Mark)**

Role of Pseudogenes (Fossil Genes):

For explaining *pseudogenes* as *evidence* of common ancestry *before* divergence. Finding the **same "fossil" gene** (pseudogene) in the **same location** in two different species is strong evidence. The "only logical explanation" is that the gene "broke" in a **common ancestor** , and both species inherited the already-broken gene before they diverged. **(1 Mark)**

36. Compare and contrast cladograms, phylograms, and chronograms. [An]

Ans: Commonality: (1 mark) All three are **types of phylogenetic trees** that show/**[CO4]** evolutionary relationships (topology).

- **Cladogram:** (1 mark) **Only shows the branching pattern (relationships)**. Branch lengths have **no meaning** and are arbitrary.
- **Phylogram:** (1 mark) Branch lengths are **proportional to the amount of evolutionary change** (e.g., number of nucleotide substitutions).
- **Chronogram:** (1 mark) Branch lengths are **proportional to actual evolutionary time** (e.g., in millions of years), often calibrated using fossil data or molecular clocks.

[4x8=32]